

Technical Task – Full Stack .NET Developer (Mid Level)

ElectroPi

Hi 🙌

Thank you for your interest in the **Full Stack .NET Developer** role at ElectroPi. As the next step, please complete the technical task below so we can evaluate your practical skills across both Backend and Frontend.

Time allowed: 2 days (48 hours) from receiving the task.

Overview

Build a simple **Task Manager** web application consisting of:

1. A **Backend API** using ASP.NET Core
 2. A **Frontend** using Angular or React (your choice)
 3. A **SQL Server** database
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Backend Requirements (ASP.NET Core + EF Core)

1. Build a **RESTful API** to manage the following entities:
 - **Project** (Id, Name, Description, CreatedAt)
 - **TaskItem** (Id, Title, Description, Status [ToDo/InProgress/Done], DueDate, ProjectId)
2. Minimum required endpoints:
 - Full CRUD for **Projects** and **Tasks**
 - Endpoint to get all tasks for a specific project
 - Endpoint to filter tasks by Status
3. Use **Entity Framework Core** (Code First) with clear Migrations.

4. Implement:

- **Data validation** (Data Annotations or FluentValidation)
 - **Proper error handling** (middleware or structured try/catch with appropriate status codes)
 - **Repository Pattern** or **Service Layer** (simplified Clean Architecture)
 - **Dependency Injection**
5. (Optional – Bonus) Structure the API so it could logically be split into independent microservices (e.g., clearly separate the Projects logic from the Tasks logic).
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Frontend Requirements (Angular / React)

1. A page listing all projects
 2. A page showing a specific project's details along with its tasks
 3. Ability to:
 - Add / edit / delete a project
 - Add / edit / delete a task
 - Change a task's status directly
 4. Use an **HTTP Client** to communicate with the API
 5. Keep the UI simple and clean (doesn't need to be highly polished — just usable and clear)
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Database

- Use **SQL Server** (or LocalDB if time is tight)
 - Design the schema so the Project–Task relationship is clear (One-to-Many)
 - Include the migration files or a schema script with your submission
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Submission

1. Push your code to **GitHub** (public repo, or private with us added as collaborators if you prefer)
2. Include a simple **README** covering:
 - How to run the project (Backend + Frontend + Database)
 - Any assumptions or design decisions you made

- Anything you'd have liked more time to complete, if applicable
3. A short demo video (2-3 minutes) showing the app running is a nice bonus, but not required.
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Evaluation Criteria

- Code quality and organization (clean code, naming conventions)
 - Understanding of architecture principles (separation of concerns)
 - Proper error handling and validation
 - Working API integrated with the frontend
 - Clarity of the README and ease of running the project
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Notes

- This task is designed to be completed within 2 days at a reasonable daily pace (not full-time), so there's no need to over-engineer or add extra features beyond what's requested.
- If you have any questions while working on it, please reach out to us right away.

Good luck!  **ElectroPi Engineering Team**